Question 1
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Flag question

Both libraries and software frameworks are collections of reusable code. Both aim to provide solutions to common problems in Software development.

Select one:
- a. Agree
- b. Cannot say conclusively
- c. Strongly disagree
- d. The correct answer is not given
- e. Disagree

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Question 7
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Flag question

Stateless is one of the six constraints in REST. Select all the statements which are true regarding this constraint.

Select one or more:
- a. Session state can be shared between the client and the server.
- b. Communication between clients and server must be stateless in nature.
- c. All of these statements are correct.
- d. None of these statements are correct.
- e. Each request from client to server must contain all the information necessary to understand the request and cannot take advantage of any stored context on the server.

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Question 2
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Flag question

What is NOT a condition that a REST API should satisfy in-order to achieive full-maturity level according to the Richardson Maturity Model?

Select one:
- a. All requests to be made using POST HTTP method in order to improve security.
- b. Proper utilzation of HTTP Verbs/ Methods and HTTP status codes.
- c. Supports Hypermedia as the engine of application state (HATEOAS) concept.
- d. None of the the given answers are valid.
- e. Individual URIs for each resource.

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Question 4
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Flag question

What is NOT a drawback of Monolithic architecture?

Select one:
- a. Lack of flexibility regarding implementation technologies.
- b. High Global complexity.
- c. Can't scale individual components.
- d. High Local complexitity.
- e. The whole application instance should be replicated if there is a need to scale.

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Question 5
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Flag question

Uniform Interface is one of the architectural constraints a REST APIs should align with. What is the main purpose of this constraint?

Select one:
- a. To clearly identify web resources through URIs.
- b. To ensure that there is no need for the client to be hard-coded with information regarding the structure of the server.
- c. All of the above.
- d. To ensure that, having accessed an initial URI for the REST application a REST client should then be able to use server-provided links dynamically to discover all the available resources it needs.
- e. To separate the client from the implementation details of the REST service using interfaces to promote independent evolvability.

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Question 6
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Select all the correct statements with regards to GitHub workflow and Git Workflow.

Select one or more:

- [x] a. Directly working on the main/ master branch is actively encouraged in Git workflow, but not in GitHub workflow.
- [ ] b. Git Workflow is very suitable for small-scale projects when compared with GitHub workflow.
- [x] c. Git Workflow is a very comprehensive workflow, whereas GitHub workflow is much more simpler that that.
- [x] d. GitHub Workflow is ideal for small-scale projects when copared with Git workflow.
- [ ] e. Git Workflow could be a better choice for managing enterprise level projects than GitHub workflow.

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Question 8
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Which of the following is TRUE regarding a feature branch in git?

Select one or more:

- [x] a. A feature branch is a copy of the main branch.
- [x] b. Ideally, the main branch should never be directly modified. Instead, the changes should first be done in a feature branch.
- [ ] c. A Git workflow like the Github workflow encourages the creation of feature branches for every feature.
- [ ] d. None of them are true.
- [x] e. A feature branch must always be created from the main branch.

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Question 9
Answer saved
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What is the purpose of a version control system?

Select one:

- a. To roll back changes to an earlier version.
- b. To keep track of changes to code or other files.
- c. To allow multiple people to work on the same file simultaneously.
- d. None of the above.
- (•) e. All of the above.

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Question 10
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What are some benefits of using an API to communicate among different software?

Select one or more:

- [x] a. APIs offer services as contracts, which other applications can consume without having to know about internal implementation details.
- [ ] b. None of these are correct.
- [ ] c. APIs abstract away the complexities of application implementation details.
- [x] d. APIs improve collaboration among developers and applications.
- [x] e. Since all APIs use JSON format to communicate, the responses are both human and machine readable.

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Question 11
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Would you recommend Microservices be used for project that is just starting out?

Select one:

- a. Cannot say conclusively.
- (•) b. No, unless there is a very specific requirement to do so, it is much better to start with Monolithic architecture.
- c. No, Monolithic architecture is much better at scaling compared to Microservices.
- d. Yes, Microservices is ideal for an application being developed by a small team of developers.
- e. Yes, Microservices should be used as they are small, independent, and loosely coupled.

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Question 12
Answer saved
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Flag question

Is it a requirement to always use JSON when sharing resource representations via REST APIs?

Select one or more:

- [x] a. No, other formats such as XML can also be used. But using JSON is a widely accepted practice.
- [x] b. It is. It is one of the six architectural constraints in REST.
- [ ] c. Plain text is also a possibility.
- [ ] d. JSON is not to be used with REST APIs.
- [ ] e. None of the above are correct.

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Question 13
Answer saved
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Flag question

Hypermedia as the engine of application state (HATEOAS) is an important concept in REST. This concept ensures that,

Select one:

- a. A client can dynamically explore the API and discover resources (Dynamic API).
- b. All the statements are correct.
- c. Having accessed an initial URI for the REST application a REST client should then be able to use server-provided links dynamically to discover all the available resources it needs.
- d. There is no need for the client to be hard-coded with information regarding the structure of the server.
- e. A client does not need any additional documentation to know the available endpoints to request resources.

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Question 14
Answer saved
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Flag question

In a regular program, the flow of control of the program is controlled by user code, which will call other library/ helper functions and methods whenever they are needed. However, in frameworks this works the other way around. It is known as,

Select one:

- a. Inversion of Control
- b. Dependency Injection
- c. Library calls
- d. Design Patterns
- e. The Singleton

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Question 15
Answer saved
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Flag question

How can you use Git branches to facilitate parallel development on a project with GitHub workflow?

Select one:

- a. By creating a new branch for each file in the project, and merging those branches back into the main branch once changes are complete.
- b. By creating a new branch for each release of the project, and merging those branches back into the main branch once the release is complete.
- c. By creating separate branches for each developer to work on.
- d. By creating a new branch for each commit in the project, and merging those branches back into the main branch once the commit is added to the staging area.
- e. By creating a new branch for each new feature or bug fix, and merging those branches back into the main branch once complete.

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Question 16
Answer saved
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Flag question

What statements below are true regarding MVC (Model-View-Controller) and MVVM (Model-View-ViewModel) architectural patterns?

Select one or more:

- [ ] a. In MVC, Model component usually handles all the data related logic.
- [x] b. In MVC, View component handles both presentation structure details as well as presentation logic.
- [ ] c. In MVVM, ViewModel handles all presentation related logic.
- [ ] d. In MVVM, View component handles both presentation structure details as well as presentation logic.
- [x] e. In MVC, Controller handles all the control related logic.

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Question 17
Answer saved
Marked out of 1.00
Flag question

In a local Git repository, the files that are ready to be committed are present in the,

Select one:
- a. Commit hash
- b. Staging area
- c. None of the above
- d. Unstaged area
- e. Working directory

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Question 18
Answer saved
Marked out of 1.00
Flag question

MVVM (Model-View-ViewModel) is a well known web frontend architecture. Select all the correct statements regarding MVVM.

Select one or more:
- [x] a. The Model component handles both the presentation structure details as well as logic.
- [ ] b. The View component is a "dumb" component (i.e. no logic, just structure for presentation).
- [ ] c. The ViewModel is the coordinator between the View and Model. It contains presentational logic.
- [x] d. The Model component contains all the data related and sometimes even business related logic.
- [ ] e. The View component handles both the presentation structure details as well as logic.

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Question 19
Answer saved
Marked out of 1.00
Flag question

According to Git Workflow, which branch should always be in a release ready state?

Select one:
- a. staging branch
- b. main/ master branch
- c. develop/ dev branch
- d. feature branch
- e. hotfix branch

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Question 20
Answer saved
Marked out of 1.00
Flag question

Layered system is one of the six constraints in REST. What is/ are true regarding this constraint?

Select one:
- a. Each layer only interacts with the layers above and below them.
- b. All of these statements are correct.
- c. The layered system constraints enable network-based intermediaries such as proxies and gateways to be transparently deployed between a client and server.
- d. The layered system constraint allows an architecture to be composed of hierarchical layers by constraining component behavior such that each component cannot "see" beyond the immediate layer with which they are interacting.
- e. Layered system constraints reduces overall system complexity as each layer only has to be aware of the layer above and below it.

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Question 21
Answer saved
Marked out of 1.00
Flag question

A software framework is an abstraction in which software, providing generic functionality, can be selectively changed (extended) by additional user-written code, thus providing application-specific software. Do you agree with above statement?

Select one:
- a. Cannot say conclusively
- b. Agree.
- c. Disagree.
- d. The correct answer is not given
- e. Strongly disagree

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Question 22
Answer saved
Marked out of 1.00
Flag question

Cacheable is one of the six constraints in REST. Select all the statements which are true regarding this constraint.

Select one or more:
- [ ] a. Cache constraint requires the data within a response to a request be implicitly or explicitly labeled as cacheable or non-cacheable.
- [x] b. If a response is cacheable, then a client cache is given the right to reuse that response data for later, equivalent requests.
- [x] c. If a response is labelled as cacheable, then the client must always store the response in its cache.
- [x] d. If a response is labelled as cacheable, then the client needs to send the said response back to the server with each request afterwards.
- [ ] e. All of these statements are correct.

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Question 23
Answer saved
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Flag question

Client-server architecture is one of the six constraints in REST. What is/ are true regarding this constraint?

Select one:
- a. It promotes low coupling between the client and the server.
- b. It promotes portability of the user interface.
- c. All of these statements are correct.
- d. It allows clients and servers grow independently.
- e. It improves scalability of the server components.

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Question 24
Answer saved
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Flag question

A merge conflict in Git is an event that occurs when,

Select one:
- a. A user tries to merge branches that have conflicting changes which git cannot resolve on its own.
- b. A user accidentally deletes a branch which was to be merged with another branch.
- c. A repository becomes corrupted and unusable, therefore preventing the users from merging any further commits.
- d. None of the above
- e. A user forgets to commit changes to a repository.

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Question 25
Answer saved
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Flag question

What is not an advantage of using software frameworks?

Select one:
- a. Community support
- b. Users of a framework should respect the limitations and conventions imposed by its design.
- c. Enables faster methods for development with less code
- d. Accelerated development
- e. All of the above.

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Question 26
Answer saved
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Flag question

Which of the following statements is false about RESTful web services?

Select one:
- a. RESTful web services can only be accessed using the HTTP GET method.
- b. All of the above.
- c. RESTful web services are stateless, meaning that each request and response is self-contained and does not rely on previous requests or responses.
- d. RESTful web services are designed to be highly scalable, as they can be distributed across multiple servers and load balanced.
- e. RESTful web services can use any data format for data exchange, including JSON, XML, and plain text.

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Question 28
Answer saved
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Flag question

Representational State Transfer Or REST is a/ an,

Select one:
- a. Architectural protocol for facilitating communications on the web.
- b. Web API protocol.
- c. Architectural style for facilitating communications on the web.
- d. Technology for creating web APIs.
- e. All of the above.

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Question 27
Answer saved
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Flag question

Which of the following is NOT an advantage of following a good Git Workflow?

Select one:
- a. Clarity in identifying the software development requirements.
- b. Faster and efficient development practices
- c. Easy team collaboration
- d. The correct answer is not given
- e. Easy source code management

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Question 29
Answer saved
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Flag question

What is/are a disadvantage(s) of using software frameworks?

Select one or more:
- [x] a. None of the given.
- [ ] b. Learning the Framework and not the programming language prevent programmers from gaining an in-depth understanding of the programming language.
- [ ] c. Users of a framework are forced to respect the limitations and conventions imposed by its design.
- [ ] d. Community support.
- [ ] e. Compliance to accepted security standards.

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Question 30
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Flag question

What is a "pull request" in Git, and how is it used in collaboration?

Select one:
- a. It is a request to merge changes from one branch into another and can be reviewed and discussed by other developers.
- b. It is a combination of git fetch and git merge commands.
- c. It is a request to delete a branch or repository and can be reviewed and discussed by other developers.
- d. It is a request to download changes from a remote repository and merge them into a local repository.
- e. It is a request to create a new branch or repository and can be reviewed and discussed by other developers.

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Question 3
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Flag question

What is/ are not an architectural constraint of REST?

1. Client-server architecture
2. Stateless
3. Cacheable
4. Strong Decoupling
5. Layered system

Select one:
- a. All of the above.
- b. 1, 2, 3 only.
- c. 1, 2, 3, 5 only.
- d. 3 only.
- e. 4 only.

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